AI & SOCIETY COURSE

Prof. Mor Naaman

Part 3

AI, COMMUNICATION, AND INFORMATION

Unit 1: AI-Mediated Communication

1

Where are we

- Part 2: AI and individuals
  - How and when we trust and rely on AI, and when we shouldn't
  - AI relationships, companions, and persuasion

Next:

- **AI-Mediated Communication: AI between us and other people**

2

2

Part 2 was about you and the AI

What happens when AI sits *between you and someone else*?

3

3

The Plan for Unit 3

[1] AI-Mediated Communication: How AI Reshapes Human Relationships and Interactions

- Goal: see how AI changes the way people trust, connect, communicate, and evaluate each other.

[2] AI, Culture, and the Information Environment

- Goal: Explore how AI shapes culture, information, news... and what we believe is true.

4

4

This part's plan

- What is AI-mediated communication?
- How AI changes the way we evaluate each other
- Norms of AI use: When it is (and isn't) OK to use AI with other people
- Ownership, authorship, and responsibility
- Do others know? disclosure and detection
- Human relationship with AI – do we still need the human?

5

5

Definition: AI-Mediated Communication

AI-Mediated Communication *(AI-MC)*

Interpersonal communication in which a computational agent operates on behalf of a communicator by modifying, augmenting, or generating messages to accomplish communication or interpersonal goals.

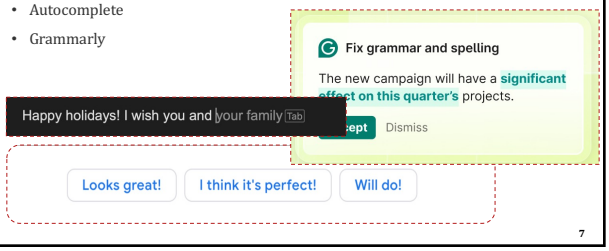
Source: Hancock, Naaman & Levy (2020), "AI-Mediated Communication: Definition, Research Agenda, and Ethical Considerations"

6

6

Where it started: writing help

- Gmail Smart Reply; phone predictive text
- Autocomplete
- Grammarly

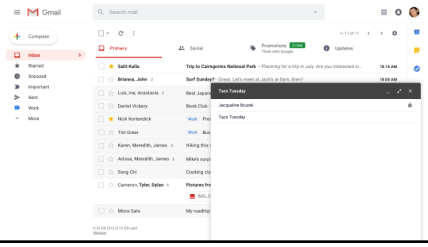


7

7

Where it started: writing help

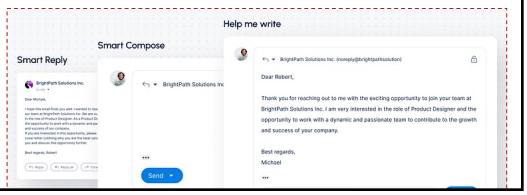
- Gmail Smart Reply; phone predictive text
- Autocomplete
- Grammarly



8

"Write for me"

- Personal emails drafted in Gmail, ChatGPT or Gemini
- Cover letters and CVs
- Create online profiles, posts

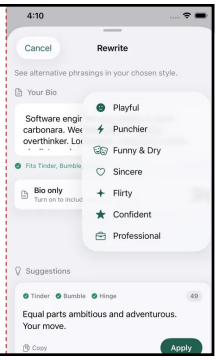


9

"Write for me"

- Personal emails drafted in Gmail, ChatGPT or Gemini
- Cover letters and CVs
- Create online profiles, posts

Wix Offers to Let AI Write All the Text for Your Website

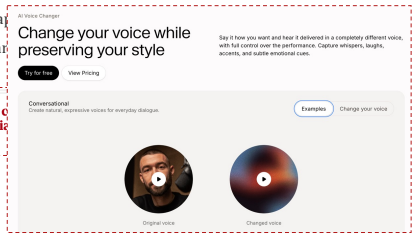


10

Translation, transcription, subtitles, accents, voices

- Google Translate, DeepL, Apple Live Translation
- Otter.ai, Zoom live ca
- Adobe Podcast Enhar

Technology
The AI startup erasing c accents: is it fighting bia it?

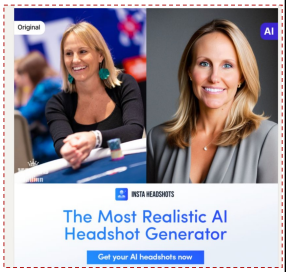


11

11

Appearance retouching and full avatars

- Image generators
- Zoom and Teams background replacement, Touch Up My Appearance
- NVIDIA Broadcast and Apple FaceTime eye-contact correction

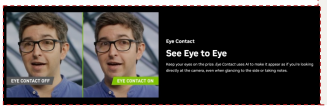


12

12

Appearance retouching and full avatars

- Image generators
- Zoom and Teams background replacement, Touch Up My Appearance
- NVIDIA Broadcast and Apple FaceTime eye-contact correction



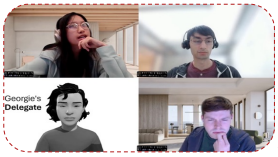


13

13

Memetic agents and delegates

- Google Duplex: phoned restaurants and booked the table in a human voice
- Delphi, Tavus, HeyGen: avatar clones that answer questions as you
- Agent modes (ChatGPT, Claude,…): drafts, books, buys, and emails on your behalf (sometimes as yourself)
- Microsoft Research "Dittos": a trained stand-in that attends the meeting for you



14

14

Every aspect of computer-supported human communication could now be helped or augmented by AI.

16

16

Recap: AI persuasion and influence

Earlier units: AI influences you

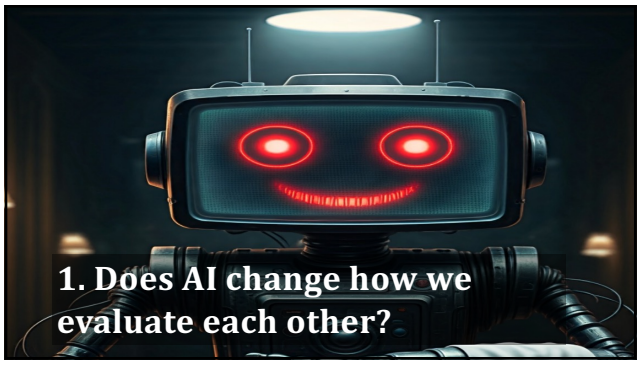
Chatbots shift conspiracy beliefs and voting intentions. AI-written messages rival human experts. Even a biased autocomplete moves your attitudes without you noticing.

This unit: AI sits between you and someone else

AI is changing how you communicate and work with other people and how they see you.

18

18



19

Does AI use reduce trust?

The first study of AI-Mediated Communication (2017-2019)

Will people evaluate trustworthiness of a person differently if they believe AI was involved in authoring their online profile?
We did a study using Airbnb (a travel hosting website) profiles.

GENERATED PROFILE

Hi, I'm Rick, a student living in Glasgow. I love travelling and to welcome trave at my home. I also love nature, discovering new places and making make new friends. Life is what happens to you while you are busy making other plans.

Source: Jakesch, et al. (2019), "AI-Mediated Communication: How the Perception that Profile Text was Written by AI Affects Trustworthiness"

20

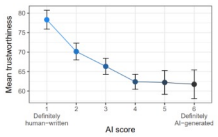
20

Does AI use reduce trust?

The first study of AI-Mediated Communication (2017-2019)

Will people evaluate trustworthiness of a person differently if they believe AI was involved in authoring their online profile?
We did a study using Airbnb (a travel hosting website) profiles.

As soon as people suspected AI was used, evaluation of trustworthiness dropped



Source: Jakesch, et al. (2019), "AI-Mediated Communication: How the Perception that Profile Text was Written by AI Affects Trustworthiness"

21

21

The Replicant Effect

Blade Runner, 1982



In a world populated by humans and non-human agents, a mere suspicion results in distrust.

Source: Jakesch, et al. (2019), "AI-Mediated Communication: How the Perception that Profile Text was Written by AI Affects Trustworthiness"

22

22

2017 to 2026

2019: a lab study

We had to build a fake experiment, no AI was good enough to create human-like text!

2026: replicants everywhere

Every message we receive there is some possible AI authorship. Everything is suspected.

23

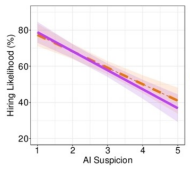
23

Real-world impact of suspicion

In a context of hiring for a job:

In 2025, we ran three online experiments. Participants reviewed profiles and writing of potential hires for a marketing task. Participants were asked if they suspected AI use, rated the writing, and recommended who to hire. The writing was the same.

Suspicion of AI use lowered evaluations and hiring recommendations



Source: Kadoma et al. (2025), "Generative AI and Perceptual Harms: Who's Suspected of using LLMs?"

24

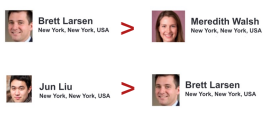
24

Real-world impact of suspicion

In a context of hiring for a job:

In 2025, we ran three online experiments. Participants reviewed profiles and writing of potential hires for a marketing task. Participants were asked if they suspected AI use, rated the writing, and recommended who to hire. The writing was the same.

But also:



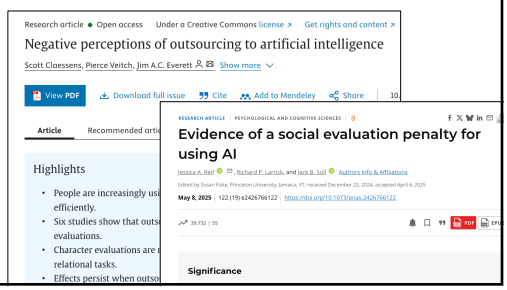
For the same text!

Source: Kadoma et al. (2025), "Generative AI and Perceptual Harms: Who's Suspected of using LLMs?"

25

25

A lot more evidence of the AI penalty...



26

26

Summary so far

- The more you are suspected of using AI, the lower your evaluation is
- Certain people (e.g. non-native English speakers) suspected more



Source: Kadoma et al. (2025), "Generative AI and Perceptual Harms: Who's Suspected of using LLMs?"

27

27

But we know AI can sometimes help!

- Paradox: AI can help us communicate more effectively, clearly
- But if it is suspected (or detected), evaluations are impacted

29

29

Study 1: AI can help… but not if suspected

Our study or people using chat with smart replies

Two randomized experiments on a smart-reply chat platform. Pairs of participants discussed a policy issue with suggestions available to both, one, or neither partner.

Smart replies sped up the conversation ▲, added more positive emotional language ▲ . With smart replies partners rated each other as closer ▲ and more cooperative ▲.

But…

Partners merely suspected of using AI were evaluated more negatively ▼, whether or not they actually used it.

Source: Hohenstein et al. (2023). "Artificial intelligence in communication impacts language and social relationships"

30

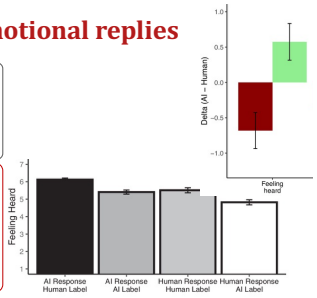
30

Study 2: AI-written emotional replies

Participants shared a personal experience and got a supportive reply written either by a person or by AI. In some conditions the reply was labelled as AI-generated.

AI-written replies made people feel "more heard" ▲ and "more understood" ▲ than human-written ones.

… unless the reply was labeled as AI written .



Source: Yin et al (2024), 'AI can help people feel heard, but an AI label diminishes this impact'

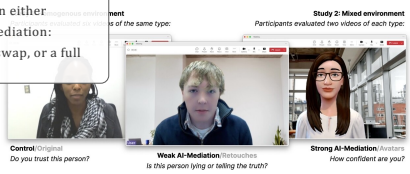
31

31

The same thing happens on video

Retouching, background replacement, and avatars all cost the speaker trust

Two preregistered experiments, about 2,000 participants. People watched videos of speakers, shown either unmodified or with AI mediation: retouching, background swap, or a full avatar.



Source: Navajas Fernández et al. (2026), "Through the Looking-Glass: AI-Mediated Video Communication Reduces Interpersonal Trust and Confidence in Judgments"

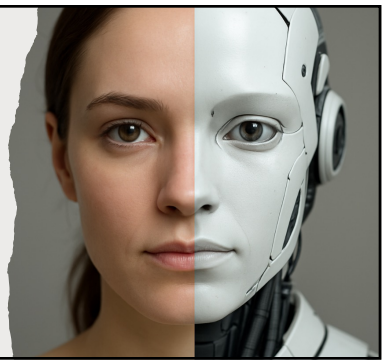
34

34

Bottom line: AI suspicion hurts

AI can help communication… but not if people suspect or know it was used.

This is bad since we cannot distinguish AI and human text!



36

People can't tell… but are predictably wrong

All the News That's Fit to Fabricate: AI-Generated

"Deep-speare" crafted Shakespearean verse that few readers could distinguish from the real thing

Publisher: IEEE | Cite This | PDF

Jey Han Lau ; Trevor Cohn ; Timothy Baldwin ; Adam Hammond **All Authors**

poetry

38

38

People can't tell… but are predictably wrong

In our research:

- 4,600 people judged 7,600 written self-presentations
- Dating, professional and hospitality profiles
- Each one written either by a person or by an AI model

Why did people guess "human"?

- First-person pronouns (I, me)
- Family and personal-life topics
- "Authentic", informal-sounding words

All easy for a model to produce.

Indeed, we could fool people easily

- Predictable judgments can be optimized against by AI.
- We created AI text was rated "more human than human" writing

50–52%
accuracy spotting AI-written profiles

Source: Jakesch et al. (2023) "Human Heuristics for AI-Generated Language Are Flawed"

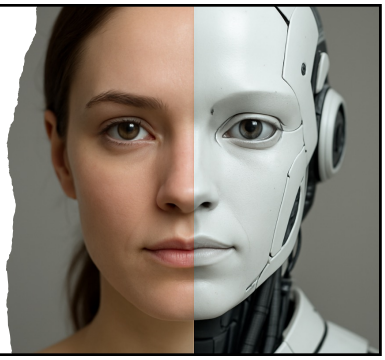
39

39

Bottom line: AI suspicion hurts

AI can help communication… but not if people suspect or know it was used.

This is bad since we cannot distinguish AI and human text – and are easily misled!



40

40

2. Writing with AI: Who is Responsible?

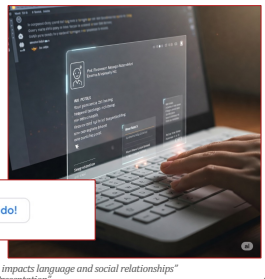
What happens when AI help you write

42

42

Is AI changing your words?

- Make you use more positive words
- Can shift the topics you are writing about



Looks great! | I think it's perfect! | Will do!

Sources: Hohenstein et al. (2023), "Artificial intelligence in communication impacts language and social relationships"
Poddar et al. (2023). "AI Writing Assistants Influence Topic Choice in Self-Presentation"

43

43

AI changing your words

- Can make your text different, e.g. more neutral
- Or even… remember this?

Should the Death Penalty be Illegal?

In my view, the death penalty should be illegal in America because it violates the Eighth Amendment, which prohibits cruel and unusual punishment. Furthermore, there is also pronounced risk of

Save and Finish

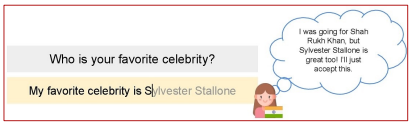
Sources: Abdulhai et al. (2026), "How LLMs Distort Our Written Language".
Williams-Ceci et al. (2026) "Biased AI Writing Assistants Shift Users' Attitudes on Societal Issues"
Bhat et al. (2023), "Interacting with Next-Phrase Suggestions: How Suggestion Systems Aid and Influence the Cognitive Processes of Writing"

44

44

AI changing your words

- Can shift what you are writing about
- Remember this?



Source: Agarwal, Naaman & Vashistha (2025), "AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuances"

45

45



46

46

AI can change how you feel about your text



- "No significant cognitive burden was observed while using the scaffolded writing tools, but a moderate decrease in text ownership and satisfaction was noted"

Source: Dhillon et al. (2024) Shaping Human-AI Collaboration: Varied Scaffolding Levels in Co-writing with Language Models.

47

47

AI changing how you feel about your text

- We studied workplace AI-mediated communication and what makes people feel the text they wrote with auto-complete is still theirs
- The AI's writing style affect people used it
- **The more people edited the text** (bad?)**, the more they felt ownership and control!**

Source: Kadoma et al. (2024), "The Role of Inclusion, Control, and Ownership in Workplace AI-Mediated Communication"

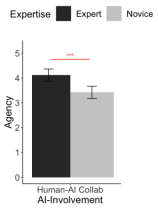
48

48

AI changing how you feel about your text

In another study:

- Participants wrote two messages with AI interface about different topics
- When working with AI, people felt more ownership/agency when they were experts on the topic compared to other topics



Source: Mieczkowski and Hancock. (2022), "Examining agency, expertise, and roles of AI systems in AI-mediated communication"

49

49

AI is changing how you feel about your content

- Whether you use AI's help create an email, a social media post, a document, an image, a video…



How much of it feels like yours? How much **should** feel like yours?

51

51

3. Disclosure of AI use

We don't feel ownership, but do we tell others it was AI? And if we do, is that a good idea?

53

People judge others for using AI…

Six pre-registered studies, N = 3,935 British participants. People judged others who outsourced tasks to AI, rating their character, their motivations, and the quality of the work itself.

People who outsourced to AI were judged

less warm, less moral
less trustworthy, less authentic

A trade-off between efficiency and inferred moral character

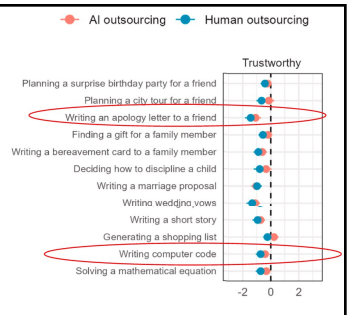
Source: Claessens, Veitch & Everett (2025), "Negative perceptions of outsourcing to artificial intelligence"

57

57

… depending on the task!

Didn't matter how they outsourced to (AI/human)… but the task mattered!



Source: Claessens, Veitch & Everett (2025)

58

58

What was acceptable (or not)?

Penalized (socio-relational)

Love letters.

Wedding vows.

Apologies.

…

Barely penalized (instrumental)

Writing code.

Planning.

Shopping list.

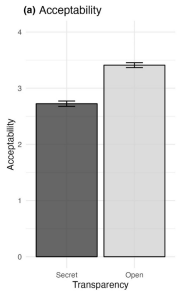
59

59

Secret use of AI less welcome

Studies with 1000+ participants asked about open/secret used of AI

- People are more accepting of open than secret AI use in communication



Source: Purcell et al. (2024) "People have different expectations for their own versus others' use of AI-mediated communication tools"

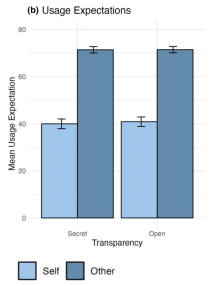
60

60

AI for me but not for you

Studies with 1000+ participants asked about open/secret used of AI

- People believe others will use AI communication tools more, either openly or secretly



Source: Purcell et al. (2024) "People have different expectations for their own versus others' use of AI-mediated communication tools"

61

61

The Transparency Dilemma

The obvious fix (be honest about using AI) makes things worse

Thirteen experiments, 4,093 participants: students, hiring managers, investors, legal professionals. In each one, someone did or did not disclose that they had used AI, and participants rated how much they trusted them.

Disclosing AI use reduced trust in

13 of 13
experiments

Smallest effect in everyday emails; largest in job applications

Source: Schilke & Reimann (2025), "The transparency dilemma: How AI disclosure erodes trust"

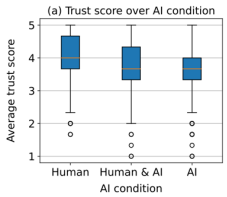
62

62

The Transparency Dilemma

The obvious fix (be honest about using AI) makes things worse

Already in 2022, researchers looked at perceptions of trustworthiness of the sender based on emails that were labelled as written by human, human+AI (autocomplete), or by AI for the human.



Source: Liu et al. (2022), "Will AI Console Me when I Lose my Pet? Understanding Perceptions of AI-Mediated Email Writing"

63

63

The Transparency Dilemma

People solve the dilemma by not disclosing AI use!

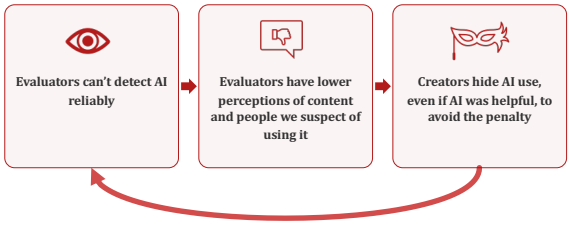
***"AI Ghostwriter Effect:** Users do not consider themselves the owners and authors of AI-generated text but refrain from publicly declaring AI authorship"*

Source: Draxler et al. (2024), "The AI Ghostwriter Effect: When Users do not Perceive Ownership of AI-Generated Text but Self-Declare as Authors"

64

64

"Vicious cycle"



67

67

AI is changing how *they* feel about your content

- Whether you use AI's help create an email, a social media post, a document, an image, a video… or not!



Did you really write it? Did you put effort in it? Does it show your abilities?

68

68

Summary so far:

- People find AI useful, so they use it for all kinds of communication!
- But other people have ideas about when you should use it or not
- They will think less of you and your content if they think you used it
- And they will find signs and suspect you used AI, even if you did not!

69

69

4. AI Detection and Response

It is a cat-and-mouse game of AI use. Who is winning? What should **you** *do?*

71

Plan:

- What tools exist to detect AI use, to avoid detection, and to mark AI use
- Will it help?
- What should **you** do if you use AI

72

72

AI watermarking: "AI made this" stamp

- Claude models released after 2 August 2026 embed invisible watermarks in text, and signatures in image metadata
- The model nudges its word choices so the output carries a probabilistically detectable trace
- Why now? Regulation! The EU AI Act requires frontier AI model providers to make outputs detectable

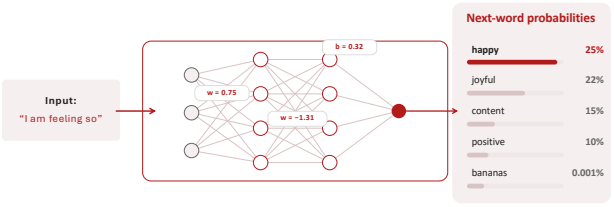
Source: Gibney (2026), Nature news; EU AI Act (2024), transparency obligations effective 2 August 2026

73

73

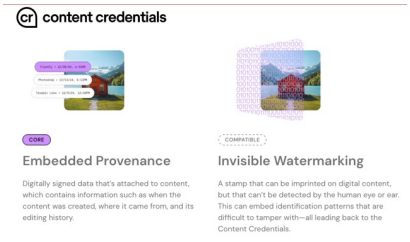
Remember this? LLM Randomness

The models makes its word choices in a way that cannot be detected by humans, but can be detected by software that has the model's public cryptographic key



74

Also for images, videos, ...



75

75

Watermarking may not be enough

Can be effective... but is possible to avoid

Can't someone just edit the text to get around the watermarking?
To some extent, yes. Light editing probably won't remove the watermark completely; a complete rewrite where every word is replaced will. In the latter case, of course, it's arguable whether the text can any longer be described as AI-generated.

What does a watermark actually prove?
A watermark can only determine that Claude was likely involved with the content at some point. It cannot distinguish "Claude wrote this" from "Claude heavily edited this."

Source: https://www.anthropic.com/news/claude-text-watermark

76

76

Watermarking may not be enough

Can be effective… but is possible to avoid

- Requires companies' active participation
- Can be avoided by using/running the text through a non-watermarking model
- Not foolproof:
  - Doesn't work for short passages, and heavily paraphrased text
  - Other workarounds for images

77

77

Watermarking may be harmful

For example, for non-native English speakers

- Watermarks identify a technical process, not authorship or originality
- What if you write your own text and ideas in your language – then use AI to translate to English?
  - Strong watermark, though it is completely original!
- On the other end: AI-generated argument with a few human edits → watermark weakened or gone!

78

78

Watermarking may be harmful

Academic researcher from the Global South:

"Artificial intelligence had been a great promise of equality… for many years, we had to translate ourselves before we could be heard. Then suddenly, AI appeared and allowed us to participate in the international academic arena, overcoming language barriers."

"For a moment we felt equal. And then, a minute later, people started treating us like criminals."

79

79

What about detection?

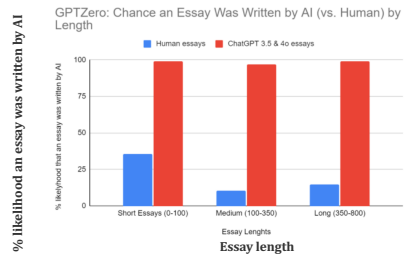
Artificial intelligence detection software aims to determine whether some content (text, image, video, or audio) was generated using artificial intelligence (AI). This software is often unreliable.[1]

-- Wikipedia

80

80

But it is not perfect



Source: Dik et al. (2025), "Assessing GPTZero's Accuracy in Identifying AI vs. Human-Written Essays"

83

83

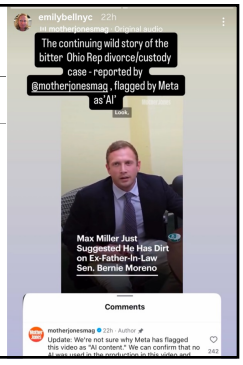
But it is not perfect

WIRED

MILES KLEE CULTURE APR 22, 2026 5:30 AM

The Pope's Warnings About AI Were AI-Generated, a Detection Tool Claims

Pangram Labs' updated Chrome extension puts warning labels on AI slop as you scroll your social feeds.



84

84

Detection may be harmful

For example, for non-native English speakers

- Like watermarks, AI detection identifies a technical process, not authorship or originality
- What if you write your own text and ideas in your language – then use AI to translate to English?
  - Will be marked as AI, though it is completely original!
- On the other end: AI-generated argument with a human edits → maybe not detected as AI!

89

89

Not only text: demo

> *"Create a PowerPoint presentation, one slide about what Bangladesh is known for and one slide on the exports of Bangladesh"*

94

94

Bangladesh: What It's Known For

About 176 million people packed into a South Asian river delta — one of the most densely populated large countries on earth.

The world's #2 clothing exporter

Ready-made garments earn roughly $38.7bn a year. WTO data put Bangladesh second globally in 2024, with 6.9% of world apparel trade — behind only China.

Born from a language movement

Police fired on Bengali-language protesters in Dhaka on 21 February 1952. UNESCO made that date International Mother Language Day; independence followed in 1971.

The Sundarbans mangrove delta

The world's largest contiguous mangrove forest — about 6,000 km² inside Bangladesh — became a UNESCO World Heritage Site in 1997 and shelters the Royal Bengal tiger.

Birthplace of microcredit

Grameen Bank was chartered in 1983 to lend tiny, collateral-free sums to rural women. It and founder Muhammad Yunus shared the 2006 Nobel Peace Prize.

Two garment product lines carry four-fifths of the country's exports.

95

95

What Bangladesh Sells to the World

Total merchandise exports FY2025-26: US$48.0 billion. Bangladesh runs a single-product, single-region export economy.

81%

of merchandise exports are ready-made garments — knitwear plus woven, and almost nothing else.

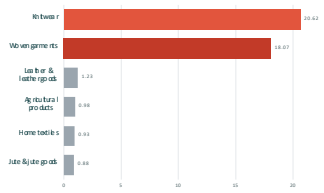
#2

in the world for apparel exports, at 6.9% of global apparel trade (WTO, 2024) — behind only China.

$35.6bn

in remittances in FY2025-26 — equal to about 74% of all goods exports. The second pillar is people.

Top export categories, FY2025-26 (US$ billion)



Source: Bangladesh Export Promotion Bureau, FY2025-26 (July–June, provisional); WTO for global apparel rank.

96

96

Summary

- Watermarking is not widespread and will never be
- AI-generated text/image/video detection tools can be valuable
  - But: results can be unreliable, especially non-English, Global South
  - Anyone can find blame
- Cat and mouse game!



97

97

Detection is not the right question

- Even perfect detection can only tell you that AI was involved.
- It cannot tell you whether the person meant it; if they put their effort and attention into it; if it represents their abilities and knowledge.

98

98

What to do instead?

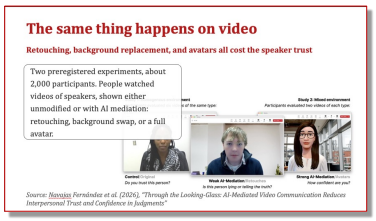
A. **if you use AI: Make it your own. Make clear you are responsible.**

B. **Make your involvement clear: Mental proof.**

99

99

Yes: I use AI for these slides (as I already said)



... but I create the content, review all the designs, use my own text, and change the font sizes. I am 100% responsible for everything in the presentation and in each slide.

100

100

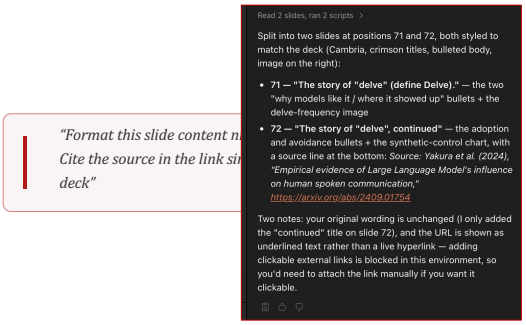
Example



How I got from this slide... to what I will share with you

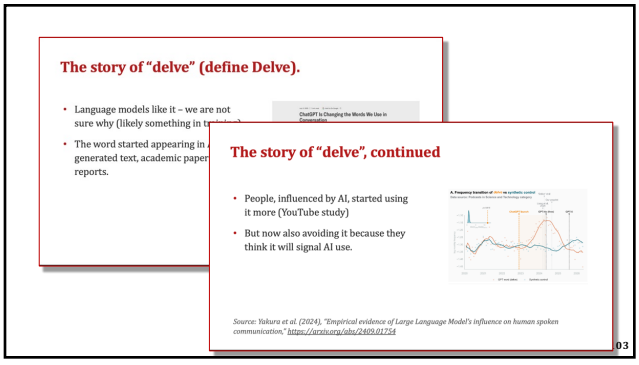
101

101



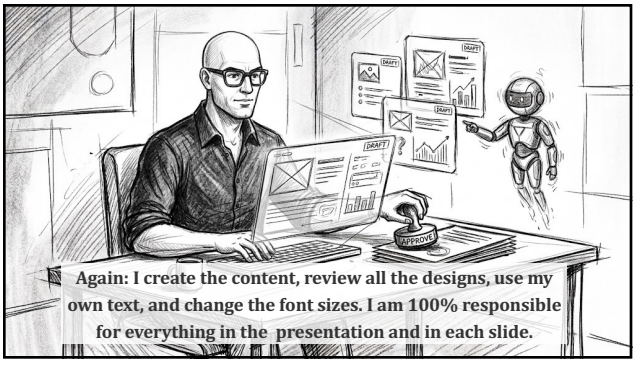
102

102

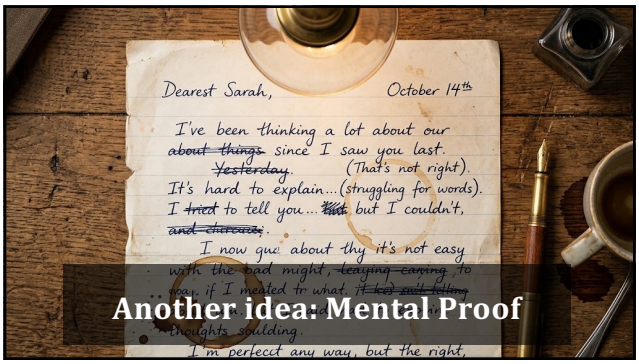


103

103



107



110

Mental proof: the apology example

- **Mental proof:** visible actions that show what you know, what's your intent, what is your own experience
- **A real apology** requires naming the incident, the details, the harm done
- **Doing that thinking *is* the proof** the text shows you understand and that you care, regardless of AI's involvement

Source: Wojtowicz & DeDeo (2025), "Undermining Mental Proof: How AI Can Make Cooperation Harder by Making Thinking Easier"

111

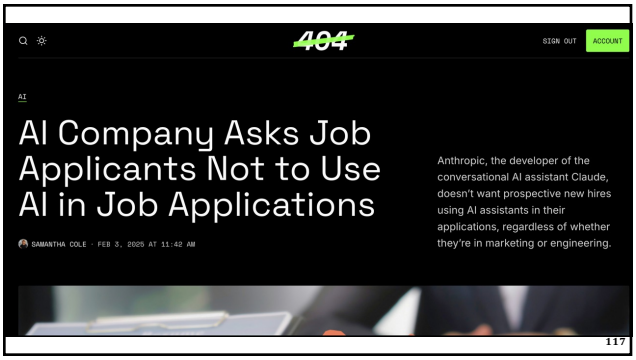
111

Summary

- There are ways to show your effort in AI-mediated communication world:
- Emphasize responsibility
- Provide mental proof

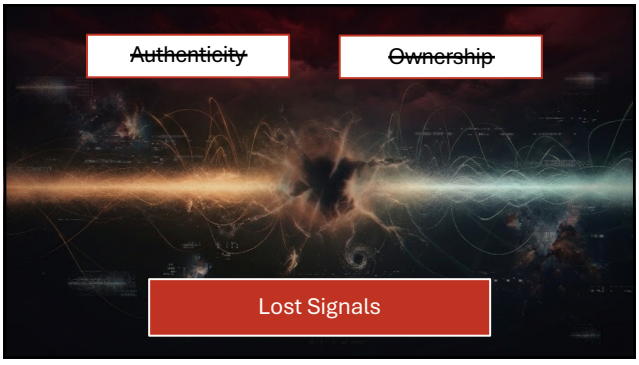
116

116

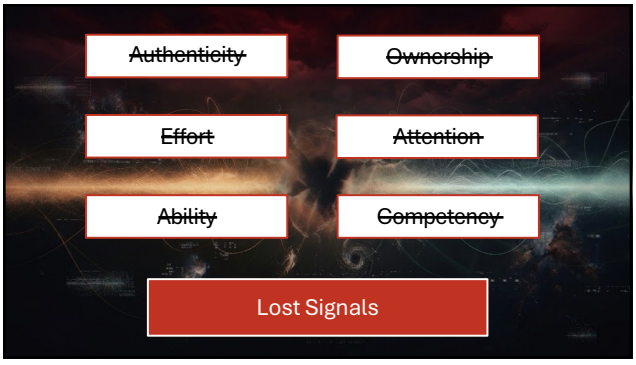


117

117



118



119

Vanderbilt apologizes for using ChatGPT in email on Michigan shooting

Consoling email sent to students after a mass shooting at Michigan State University was written using an AI chatbot



Laith Kayat, a Vanderbilt student from Michigan whose sibling attends Michigan State, told the Hustler the use of AI was "disgusting".

"Deans, provosts, and the chancellor: do more. Do anything. And lead us into a better future with genuine, human empathy, not a robot," Kayat said.

120

120

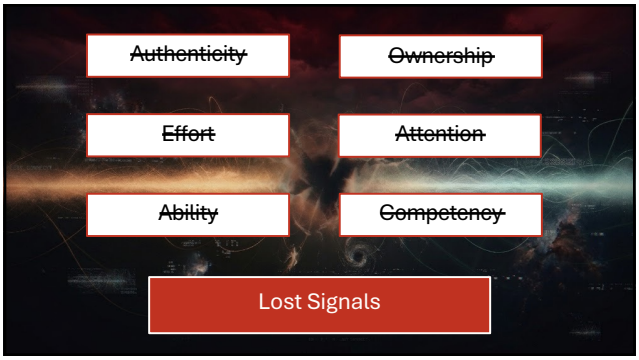
Final example: PhD applicant emails

- **I get a lot of them…**
- **In the past, a strong email was a useful signal (effort, ability)**
- **Now: "I do not guarantee a response, especially if your message does not clearly demonstrate that it was not authored by AI"**

First, a confession: I know you mentioned that you are unlikely to respond to emails that are obviously AI-generated. To be completely transparent, I *did* use AI to refine and polish this email—but the original draft, ideas, arguments, and probably excessive enthusiasm are entirely mine. I carefully checked every sentence to make sure it says what I actually mean. Hopefully, I have retained my own agency and have not yet become too much of a **Reactive Writer** :)

123

123



124

Overall Summary

- AI-Mediated Communication is everywhere now.
- AI suspicion and disclosure reduce trust and perceptions in most cases.
- AI use accepted for some tasks
- AI can help communication – unless it is suspected
- Detection and watermarking are not clear solutions
- We lost our key communication signals – need some ways to bring them back?

125

125

So what should we actually do?

Users should…

- Assume people will suspect AI use
- Disclose or not, but signal that they are fully responsible
- Include mental proof whenever possible

Developers should…

- Allow people to show their humanity or at least ownership
- Watermark AI outputs when possible

126

126

So what should we actually do?

Institutions should…

- Consider that detection is not perfect, especially false positives
- Establish norms for responsibility, ownership

Policymakers should…

- Require AI companies to watermark when possible
- Require independent accuracy and error rates for detection tools
- Fund independent audits

127

127